

QPC module

Audit client

Cloud provider

QPU

1. self-tests; instantiate DRBG

2. $s \leftarrow \text{Generate}(256 \text{ bits})$

3. $E_s = \text{HSE}(s, d); J = E_s \circ \text{logical circuit};$
 $c = \text{SHA3}(s \parallel r)$

4. $\sigma_C \leftarrow \text{ML-DSA.Sign}(J \parallel c)$

5. hybrid X25519 + ML-KEM-768 $\Rightarrow K;$
 AES-256-GCM $_K(J, c, \sigma_C)$

6. compile, execute J 7. counts, syndromes R

8. R , calibration snapshot cal,
 $\sigma_P \leftarrow \text{ML-DSA.Sign}(H(J) \parallel R \parallel \text{cal})$

9a. $\text{Verify}(\sigma_P)$

9b. reveal (s, r) ; recompute E_s ;
 audit accepted logical disturbance